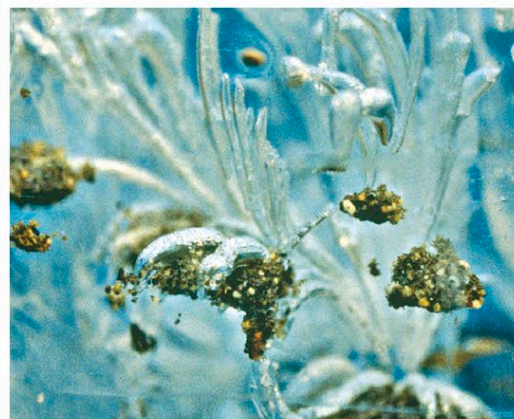




< EVIDENCE OF CLIMATE CHANGE

Studies of polar ice core samples have revealed details of past climate change. The chemical analysis of trapped gas bubbles helps provide an approximate measure of the temperature of the atmosphere at the time the ice was formed.



< ICE CORE SAMPLE

This is a close-up of an ice core sample recovered from Lake Bonney, Antarctica, which has a permanent ice cover. It shows trapped bubbles of air and sediment particles from the lake bed.

CLIMATE CYCLES

There is clear evidence from rocks and fossils that Earth's climate has changed significantly over time and that this has affected the evolution and distribution of life, and led to the extinction of many species. There are a number of causes of this natural climate change, including volcanic activity – which pollutes the atmosphere with ejected gas and dust – and changes in ocean currents, which move heat around the globe. Change is also driven by the orbital and rotational cycles of the planet, which affect the amount of solar radiation that reaches Earth's surface. This in turn influences the planet's temperature and climate, and triggers Earth's cycles of ice ages and greenhouse periods.

CHANGING GEOGRAPHY

Over time, continents have been displaced as oceans have expanded and contracted due to plate tectonics. As continents moved from one hemisphere to another they passed through different climatic zones and, at times, formed supercontinents. The

size of these huge landmasses affected regional climates. In addition, the changing shape of ocean basins has altered water circulation, which in turn changes the temperature and humidity of the atmosphere above, and so affects climate.

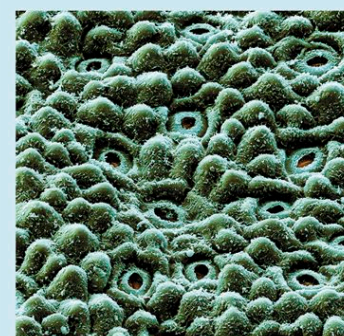
GREENHOUSE AND ICE AGES

Long-term changes in climate are divided into cold periods – ice ages, when there were long-lasting ice sheets at the poles – and warmer, greenhouse periods with largely ice-free poles. Warm phases are linked to the release of greenhouse gases, such as carbon dioxide, by plants into the atmosphere. These trapped heat in Earth's atmosphere and produced huge, shallow seas, arid zones, and lush forests that, in the time of the dinosaurs, provided them with plentiful food. Ice ages, lasting millions of years, can be traced from the impact that glaciation has left on the landscape. Fossils show how rapid climate change associated with ice ages had a dramatic impact upon life on a global scale.

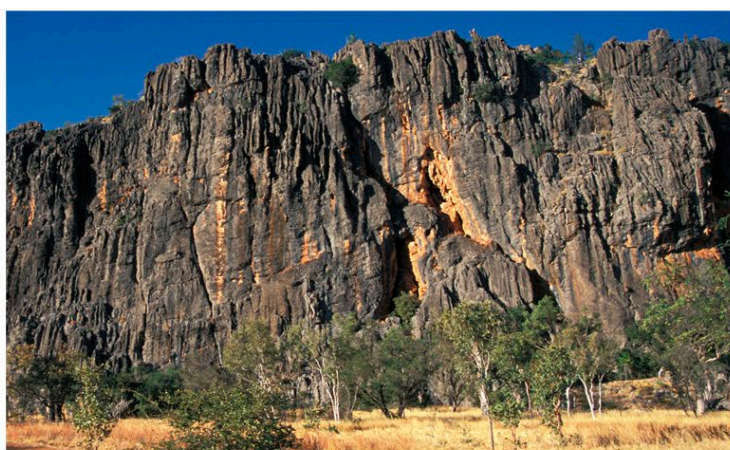
SCIENCE

STUDYING STOMATA

Plant growth depends upon an exchange of gases between the atmosphere and the plant tissue, through special gaps between the leaves' cells called stomata. The stomata open and close to take in carbon dioxide for photosynthesis and allow waste water and oxygen to escape. In general, plants adapt to high levels of atmospheric carbon dioxide, associated with warm "greenhouse" climates, by evolving high densities of stomata on the leaf surface. Fossil evidence of changes in the stomatal density of certain plants tracks changes in atmospheric carbon dioxide over time.



EUCALYPTUS STOMATA



^ DEVONIAN CORAL REEF

Earth's changing climate is spectacularly illustrated by this limestone outcrop in The Kimberleys, W. Australia. In the Devonian Period (400 million years ago), the area was under water and the cliff was a barrier reef.

✓ CARBON DIOXIDE AND TEMPERATURE

Gas bubbles trapped in polar ice cores indicate the fluctuating temperature of the planet – the higher the amount of carbon dioxide detected in the ice core, the higher the atmospheric temperature would have been at that time.

